

Week 6 FinTrust Security Architecture Summary

High Availability (HA) Layer: FinTrust's architecture is built on a resilient high-availability foundation that distributes workloads across redundant infrastructure components, ensuring banking services remain available during component failures, maintenance activities, or traffic spikes. Load-balanced application services, fault-tolerant networking, and resilient cloud resources provide continuous service delivery, creating a stable platform upon which identity, security, and monitoring capabilities operate.

Identity and Access Management (IAM) Layer: The IAM layer enforces controlled access for employees, customers, and administrators. Employees authenticate through IAM Identity Center integrated with Active Directory, enabling centralized single sign-on without individual IAM users. Customers authenticate through Amazon Cognito and receive temporary, least-privilege credentials, while DevOps administrators operate within permission boundaries that prevent privilege escalation. Together, these controls ensure that only authorized users receive access appropriate to their roles.

Security Services Layer: Defense-in-depth is provided through AWS Config, Secrets Manager, ACM, AWS WAF, Shield Standard, and Systems Manager Session Manager. AWS Config continuously validates compliance, Secrets Manager automates credential rotation, ACM manages trusted TLS certificates, WAF blocks common web attacks such as SQL injection and cross-site scripting, Shield Standard provides DDoS protection, and Session Manager replaces traditional bastion hosts with audited, IAM-controlled administrative access. This layer strengthens governance, reduces attack surface, and supports regulatory compliance objectives.

Threat Detection and Response Layer: Continuous monitoring and automated incident response are delivered through GuardDuty, EventBridge, Lambda, CloudTrail, VPC Flow Logs, SNS, Amazon Detective, and forensic snapshot capabilities. High-severity GuardDuty findings trigger automated workflows that isolate affected EC2 instances through restrictive security groups, notify security teams, preserve evidence through EBS snapshots, and enable investigation in Amazon Detective. CloudTrail data events and VPC Flow Logs provide visibility into user activity and network traffic, allowing rapid detection, containment, and analysis of suspicious behavior.

CISO Perspective: These four layers operate as a unified security architecture: the HA layer keeps critical financial services available, the IAM layer validates who can access resources, the security services layer protects workloads and enforces compliance, and the threat detection layer identifies and responds to malicious activity. Together they provide a resilient, auditable, and security-first environment aligned with FinTrust's operational resilience and risk management objectives.